Contoso Health readmissions readout

What drives the HRRP penalty, what next cycle looks like, and what would reduce it

2026-10-21

Contoso Health is a fictional client. Its 12 facilities are real public Ohio CMS Certification Numbers relabeled with invented names, and its operational data are synthetic. No patient data (PHI) is used anywhere (`engagement/phi-note.md`).

Every number in this deck is listed with its source on the last slide.

The question

> CMS penalizes us under the Hospital Readmissions Reduction Program. Which of our facilities and conditions drive the penalty, what would actually reduce it, and does the penalty itself change hospital behavior?

Restated so the data can answer it (`engagement/discovery.md`):

1. Which facilities and conditions are above the cutoff now, and which will be next cycle?
2. Does anything Contoso does lower the excess readmission ratio (ERR)? Candidate: follow-up calls after discharge. This needs an experiment, not CMS data.
3. Does being counted as penalized change a hospital's later ERR?

The ERR compares a hospital's readmissions with what CMS expects for its patients: 1.02 is 2 percent more than expected. A condition above the cutoff counts toward the payment cut.

The data

Source	What it gives
CMS HRRP, FY2013 to FY2027	ERR and discharges per hospital and condition, 266,159 rows
CMS Hospital General Information	Attributes of 5,419 hospitals
Census ACS 5-year, 2011 to 2024 vintages	County income, poverty, insurance, age, disability
Contoso operational database (synthetic)	12 facilities, 1,850 monthly discharge rows, 3,700 follow-up call rows
MTSamples	2,357 public clinical notes standing in for Contoso's notes

Two limits shape every answer:

- **The cutoff is CMS's published value.** 1.0 for FY2013 to FY2018, each year's peer group median from FY2019. 1.0 stands in only where CMS published no median: FY2023 pneumonia, which CMS left out of that year's program, and hospitals missing from a year's file.
- **Dollars are estimates.** CMS publishes each hospital's payment adjustment factor but not the base payments it applies to; the base is estimated from CMS's own IPPS impact file (`docs/penalty-dollars.md`).

What drives Contoso's penalty

FY2026 ERR by facility and condition (bold: above that year's cutoff, the peer group median)

Facility	AMI	HF	PN	COPD	Hip/knee	CABG
Alder	0.961	0.952	0.897	**1.008**	**1.074**	**1.043**
Birch	**1.131**	**1.077**	**1.090**	0.996	**1.161**	
Dogwood	**1.184**	**1.021**	**1.065**	0.993	**1.190**	0.970
Hawthorn	**1.068**	**1.044**	**1.025**	**0.999**	**1.630**	**1.061**
Oak		**1.128**	**1.087**	**1.006**	**1.006**	

The other seven facilities (Cedar, Elm, Fir, Juniper, Linden, Maple, Spruce) have 12 pairs above the cutoff between them, at most three each. 33 Contoso pairs were above the cutoff in FY2026. Peer medians sit just below or above 1.0, so a ratio of 0.999 can count (Hawthorn COPD, against a median of 0.9935).

- **Hip and knee replacement is the widest driver:** 8 of the 9 Contoso facilities that report it are above the cutoff. Nationally hip/knee also has the widest spread of any condition.
- **Four facilities carry most of it:** Birch, Dogwood, Hawthorn and Oak are above the cutoff on four or more conditions each, 18 of the 33 pairs. Hawthorn's hip/knee ratio, 1.630, is the highest of any hospital in the country that year.

Where Contoso stands now (FY2027, exact cutoff)

- **10 of 12 facilities** have at least one condition above the cutoff; **25 of 67** reporting facility-condition pairs are above it. Nationally, 90.7 percent of hospitals had at least one condition above the cutoff in FY2027.
- **Contoso's mean ERR is 0.999**: 0.0092 below Ohio's mean and 0.0029 below the national mean.
- Contoso's mean was above the national mean in FY2015, FY2016 and FY2023 to FY2026, and above Ohio's in FY2023 to FY2026.
- **Against other Ohio hospitals, Contoso's median hospital is 0.0055 lower** (FY2026, adjusted p 0.87). With 12 hospitals that is not distinguishable from no difference, which is not the same as evidence that Contoso is like its peers.

Reading: Contoso is a typical Ohio network. Its exposure comes from a few facilities and from hip/knee, not from the whole network running high.

What the penalty costs, and which year can still change

The fiscal year names the year the penalty is paid, not the discharges it measures.
Every dollar figure is an estimate: (1 - CMS's payment adjustment factor) x base operating DRG payments estimated from CMS's IPPS impact file and Table 1B.

Fiscal year	Paid	Discharges measured	Where it stands	Estimated penalty
FY2026	Oct 2025 to Sep 2026	07/2021 to 06/2024	The bill paid; comparison year	about $2.0M
FY2027	Oct 2026 to Sep 2027	07/2023 to 06/2025	The bill being paid now; published Aug 2026, fixed	about $1.8M
FY2028	Oct 2027 to Sep 2028	07/2024 to 06/2026 (assumed)	Mostly baked in; the model's forecast (next slide)	not estimated
FY2029 on	Oct 2028 on	07/2025 on (assumed)	What Contoso can still change	see the pilot economics

- **What changed, FY2026 to FY2027 (down about $195K):** AMI fell $447K, most of it Dogwood ($373K to $13K); Birch hip/knee fell $150K. CABG rose $530K: Alder ($57K to $364K) and Hawthorn ($52K to $238K); Oak heart failure and hip/knee rose $160K together.
- A program started in 2027 first moves the penalty around FY2029 to FY2031.

Next cycle: what the model expects for FY2028

The penalty model gives, for every hospital and condition, the probability that the ERR lands above the cutoff next year. It is a ranking tool; it does not predict the ERR or the dollars.

- **29 Contoso pairs** have a probability of 0.5 or more for FY2028, against 25 above the cutoff in FY2027.
- Highest: **Oak hip/knee (0.995), Oak heart failure (0.994), Hawthorn CABG (0.987), Alder CABG (0.985).**
- What it leans on: last year's ERR dominates, then last year's above-cutoff flag, the mean prior ERR, the last ERR change and state. County socioeconomics add little.

Use: start each cycle's review with the pairs at the top of this list, and with Birch, Dogwood, Hawthorn and Oak, which carry most of the current exposure.

How good the model is, and where it is weak

Held-out year	Carry last year's flag forward	Logistic regression	LightGBM (chosen)
FY2026	AUC 0.767	0.863	**0.868**
FY2027	AUC 0.635	0.676	**0.679**

- AUC is the chance the model ranks a pair that went above the cutoff over one that did not; 0.5 is a coin toss. LightGBM's lead over logistic regression is small; the larger step is from the carry-forward rule to either model.
- **FY2027 is the harder test.** Its performance window overlaps FY2026's by one year instead of two, so last year's ratio says less. FY2028 follows FY2027 and is more likely to look like the 0.679 than the 0.868. Labels use CMS's published cutoff in every year.
- **Overconfident on FY2027:** pairs given under 0.1 were above the cutoff 21 percent of the time, pairs given over 0.9 only 75 percent. Read the probabilities as an order, not as odds.
- A gate stops a new model from replacing the current one if its test AUC is below 0.80 or below logistic regression's.

Does the penalty change hospital behavior? No.

Regression discontinuity at the 1.0 cutoff, FY2013 to FY2018 (the pre-registered sample), outcome the change in ERR to the next non-overlapping three-year window. Design fixed before outcomes were examined (`docs/analysis-plan.md`).

Measure	Result
Estimate (jump in ERR change at 1.0)	+0.0004
95% interval (robust, bias-corrected)	-0.0027 to +0.0047
Smallest effect the data could detect (80% power)	0.0053
Bandwidth	0.061 either side; 0.5x to 2x all include zero
Placebo cutoffs at 0.90, 0.95, 1.05, 1.10	no jump at any (adjusted p 0.61 to 0.75)
Sample	69,163 rows, 3,268 hospitals
Added after pre-registration: FY2013 to FY2025 at published cutoffs	+0.0002 (-0.0023 to +0.0032), the same null

A precise null. Hospitals pushed just over the line and publicly counted as penalized ended the next window with the same ratio change, within a few tenths of a percent, as hospitals just under it. Effects larger than 0.02 either way are ruled out; an effect a quarter that size would have been seen.

What the causal result does not show

- **Hospitals far above the line.** The design compares hospitals near 1.0, where the payment cut is small. A response to larger cuts is not ruled out.
- **A retention caveat.** A density test on every row rejects (p 0.009): fewer rows sit just above 1.0 than just below. In the analysis sample it does not (p 0.91), but rows just above 1.0 are 3.5 points more likely to have a later ERR at all. That could bias the estimate by an unknown amount; bounds have not been run.
- **The difference-in-differences check disagrees in sign.** It finds a small improvement, -0.0030 (-0.0058 to -0.0002), where the RDD finds +0.0004. Intervals overlap, but by the pre-registered rule it does not support the RDD. Both are under a third of the 0.02 threshold.

For Contoso: the penalty label alone will not lower a facility's ratio. And a penalized facility's ratio falling next cycle is not proof a program worked: ratios near and above 1.0 drift back toward 1.0 on their own. Programs need a comparison group, which is what the pilot provides.

What the notes add

Clinical notes hold risk factors CMS data does not: comorbidities, prior admissions, discharge disposition, social risk, follow-up plans. A model hosted inside the Databricks workspace reads each note and returns those fields, each with a quote from the note; nothing leaves the workspace.

	gpt-oss-120b (chosen)	llama-3.3-70b	llama-3.1-8b	gemma-4-26b (local)
Mean field F1 on 50 labelled notes	**0.81**	0.70	0.54	**0.85**
Rejected after one retry	4 percent	4 percent	28 percent	0 percent

- **Label review:** the project lead drafted the labels and checked all 50 against the written rules; Chris Teceno, a clinician, reviewed all 50 and decided the four judgment calls; one label changed and the ranking held. With 4 primary-condition positives, one note moves a field's precision or recall by 0.25 or more.
- **Local option:** the fourth column is a model run on a laptop, so no note text leaves the machine. On these notes it scored no lower than the hosted model, which makes keeping notes inside Contoso's network a hosting question, not an accuracy one. It is not in the pipeline,

Recommendation: test follow-up calls in a stepped-wedge pilot

A structured nurse call within 72 hours of discharge for HRRP conditions, switched on two facilities at a time (one larger, one smaller) every three months in an order drawn by lot.

Item	Design
Facilities	12, each compared with itself before and after its switch
Duration	21 months of enrollment, answer one month later
Sample	about 28,672 discharges (synthetic Contoso volumes)
Usual-care readmission rate	16.9 percent
Detectable at 80% power	a drop of 3.44 points (to about 13.5 percent)
Power for a 2-point drop	38 percent

- 3.44 points is a large effect for a phone call. If the calls work as hoped, the pilot sees it; if they work modestly, it will say the effect is under about 3.5 points, not whether it is zero.
- Decisions are fixed now: a clear drop with no rise in ED visits and staff hours within budget makes the calls standard; a result that rules out a 2-point drop stops the script.
- Design under review in PR #13 (`docs/pilot-design.md`).

Would follow-up calls pay for themselves?

Contoso's own ledger: FY2027 penalty avoided minus the program's cost. Estimates; costs are BLS wages plus stated time, setup and management assumptions (central $525K a year, range $295K to $1.01M, for about 16,400 discharges a year).

Effect on the 30-day rate	Basis	Penalty avoided	Contoso net, central cost
0 points	Pooled RCTs of calls alone (Boggan 2025)	$0	-$525K
1.0 point	A judgment	$0.50M to $0.93M	-$22K to +$404K
3.0 points	Multicomponent bundles (Leppin 2014), not a call alone	$1.20M to $1.64M	+$679K to +$1.12M
3.44 points	The pilot's detectable effect; more than the evidence supports	$1.29M to $1.71M	+$770K to +$1.19M

Ranges are 0.5 to 1.0 of the drop reaching each ERR.

- **Trials find no effect from a call alone. The program pays for itself at Contoso at about half a point to a point** (0.53 to 1.05 points at the central cost), which the pilot cannot detect; at the 1-point judgment it about breaks even.
- Medicare and insurers would save about $14,000 per avoided readmission ($2.30M a year at 1 point). That is their money, never added to Contoso's.
- Contoso's upside is capped by the penalty it pays, about $1.75M a year.

Limits, stated plainly

- Contoso is fictional. Its CMS numbers are real public data for relabeled Ohio hospitals; its operations (volumes, follow-up calls) are synthetic and cover only FY2026 and FY2027.
- Cutoffs are CMS's published values; 1.0 stands in only where CMS published no peer median (FY2023 pneumonia, and hospitals missing from a year's file, about 200 rows a year).
- The model is a ranking with AUC 0.679 on FY2027 and is overconfident there.
- The causal null applies to hospitals near the cutoff, and carries the retention caveat.
- Penalty dollars are estimates: the base payments are modelled from CMS's impact file, not Contoso's paid claims. Program costs rest on stated assumptions, and the 1-point central effect is a judgment.

Next steps and decisions needed

1. **Use the FY2028 ranking** in the quality review: Oak, Hawthorn, Birch and Dogwood first, and hip/knee across the network. The dashboard and Genie space answer the five agreed questions.
2. **Approve the pilot**: confirm staged rollout is possible, which facilities cannot be randomized, and the 2-point smallest reduction worth the program (finance).
3. **Azure:** the Key Vault and Data Factory run is done (evidence); the resources are torn down after the readout.
4. **Ask the DBA** whether the operational database sets `updated_at` on update; the incremental load depends on it.

Sources

All files are on `main` unless marked; Gold tables and notebook runs are from 2026-10-21, the penalty dollar tables from 2026-10-26, the nurse-call economics and pilot power were rerun on 2026-10-28 on corrected synthetic volumes, and the cutoff, model and causal figures are from the 2026-10-27 and 2026-10-28 runs that loaded the FY2019 to FY2026 peer medians (ADR 0008).

- `engagement/discovery.md` (Data available): 266,159 HRRP rows; 5,419 hospitals; ACS 2011 to 2024 vintages; 12 facilities, 1,850 discharge rows, 3,700 call rows; 2,357 notes.
- `docs/gold.md` (Penalty cutoff) and `engagement/adr/0008-published-peer-median-cutoffs.md`: 1.0 to FY2018, the published peer median from FY2019; the rows still on 1.0 (FY2023 pneumonia, hospitals missing from a year's file, about 200 rows with an ERR a year).
- `gold_contoso_scorecard`, FY2026, queried 2026-10-28: 33 pairs above the cutoff, 18 at Birch, Dogwood, Hawthorn and Oak, 12 at the other seven; Hawthorn COPD 0.999 against a peer median of 0.9935; hip/knee above the cutoff at 8 of 9 facilities.
- `docs/eda/01_eda.html` (notebook export, section 5, Contoso condition table, CROSS_SECTION_YEAR 2026): every FY2026 facility ERR on slide 4; Hawthorn hip/knee 1.630 (also the FY2026 hip/knee maximum in section 2); D11 difference in medians -0.0055, BH p 0.868. Summarised in `docs/eda/what-the-data-can-answer.md`: hip/knee 8 of 9, widest spread (3.4 times COPD's), 90.7 percent penalized in FY2027.
- `docs/genie.md` (Q1 and reworded check): FY2027 10 facilities, 25 of 67 pairs, mean ERR 0.999; FY2026 31 pairs under the earlier 1.0 stand-in (33 now). (Q2): FY2027 mean 0.9992, 0.0092 below Ohio, 0.0029 below national; years above national and Ohio. (Q3), for model version 4: 28 pairs at 0.5 or more; the version 7 figures on slides 7 and 8 are in `docs/model-card.md`. Facility ids to names: `src/readmission/ingest/contoso.py`.
- `docs/model-card.md` (Metrics, FY2028 scores, Calibration, What the model uses, Metric gate; MLflow experiment `/Shared/readmission-penalty`, run `f724b6647aa94a2c865810c96846ec4b`, job run 1074163289288573, model version 7): AUC 0.767, 0.863, 0.868 (FY2026) and 0.635, 0.676, 0.679 (FY2027); 29 Contoso pairs at 0.5 or more, Oak (CH11) hip/knee 0.995 and HF 0.994, Hawthorn (CH07) CABG 0.987, Alder (CH01) CABG 0.985; 21 and 75 percent calibration; 0.80 gate.
- `docs/causal-findings.md` (The answer; Validity checks; Sensitivity; Difference-in-differences; Limitations), from `docs/causal/02_causal.html`: +0.0004, -0.0027 to +0.0047, MDE 0.0053, bandwidth 0.061, 0.5x to 2x, threshold 0.02, 1.02 read as 2 percent more readmissions than expected, placebo cutoffs 0.90, 0.95, 1.05 and 1.10 with BH p 0.61 to 0.75, 69,163 rows, 3,268 hospitals, density p 0.009 and 0.91, retention 3.5 points, DiD -0.0030 (-0.0058 to -0.0002); secondary sample +0.0002 (-0.0023 to +0.0032), MDE 0.0039, 138,043 rows.
- `engagement/adr/0002-llm-risk-factor-extraction.md` (Results; First job run; MLflow runs rescored on the reviewed labels `e18026f5a77b444db0ecc9fb8163c8bf`, `321fafc2509d4fc09edc42476355f193`, `0fe0fa0c99ce40e4a4232a139c9ecd70`): F1 0.81, 0.70, 0.54; reject rates 4, 4, 28 percent; 4 primary-condition positives, 0.25 swing; 100 notes, 14 rejected. `docs/llm-extraction.md`: all 50 labels drafted and checked by the lead, reviewed by Chris Teceno, a clinician, who decided the four judgment calls; one label changed.
- `docs/pilot-design.md` (on branch `pf/pilot-experiment-design-post-discharge-follow-up`, PR #13, open; Summary and For the COO; `analysis/03_power.py`, seed 20261021): 72-hour call, 12 facilities, 6 steps of 2 every three months, 21 months plus one month, about 28,672 (28,700) discharges, 16.9 percent usual care, MDE 3.44 points to about 13.5 percent, 38 percent power at 2 points, smaller than about 3.5 points, 2-point decision threshold. Rerun 2026-10-28 on the corrected volumes (all-payer factor 3.2): MDE 3.82 before, 3.44 after; `engagement/status-update.md` keeps the 3.82 it reported on 2026-10-21.
- `docs/penalty-dollars.md` (Headline; Timeline; What changed; Economics; Break-even), from `gold_penalty_dollars`, `gold_penalty_by_condition`, `gold_penalty_sensitivity` and `gold_nurse_call_economics`: $1,959,057 (FY2026) and $1,764,148 (FY2027); FY2027 windows and publication; condition and facility changes; costs $295,192, $524,714, $1,005,542 for 16,400 discharges a year; penalty avoided $503,059 and $928,851 at 1 point, $1,203,846 and $1,642,249 at 3, $1,294,498 and $1,710,033 at 3.44; net at the central cost -$21,655 and +$404,138 at 1 point; break-even 0.53 and 1.05; payer savings $2,296,000 at 1 point; $14,000 per readmission (Medicare $13,986 per Contoso discharge, DY2024); Boggan 2025, Leppin 2014.
- `docs/synthetic-data.md` (How it is generated; Realism check; Finding): how Contoso's synthetic volumes and calls were made, seed 20261021; all-payer factor 2.0 against a public estimate of 3.21, so synthetic volumes ran about 38 percent low; corrected to 3.2 on 2026-10-28 (Finding), which moved the pilot MDE from 3.82 to 3.44 points and the central program cost from $371K to $525K.
- `engagement/status-update.md` (Next; Decisions needed): pending Azure run and screenshots; the four decisions.